Pictures that fool a model

AI security · Level 3 · Lesson 3



Today we will

- Explain how small, chosen changes fool a model.
- Test whether an answer is fragile with small nudges.
- Decide where a person must check a model's answer.

Remember

- Adversarial examples: changed on purpose, invisible to us, wrong for the model.
- A model can be sure and wrong: confidence is not a guarantee.
- Train it tough, test it, and keep a person deciding what matters.

Fool the sorter

1. Warm-up (5 min): How do you tell a cow from a goat?
2. Teach (8 min): Shortcuts, and small changes that fool models.
3. Fool the sorter (15 min): Find the smallest change that flips it.
4. Toughen it (7 min): Rewrite the rule and test again.
5. Wrap-up (5 min): Where must a person check? Then the exit ticket.

Make the secret rule depend on a tiny detail (a dot in one corner, a line slightly too long), just as models latch onto small details. Keep the drawings simple so every group can play.

Let’s discuss

1. Why could a tiny change fool our “model” but not us?
2. If a model is 99% sure, should we trust it? When?
3. Where should a person always have the final word?

Exit ticket

- Write one place where an AI's answer should never decide alone, and who should check it.